

Multi-Agent Collaboration & Swarm Intelligence.

The canonical architecture for how every intelligent agent inside Auxion collaborates. AIS-001 defined the cognitive subsystems of a single reasoning engine; AIS-002 defines the autonomous consulting workforce — a swarm of thirty single-responsibility agents that analyze a business the way an elite firm would, held to deterministic, explainable engineering discipline. It refines and extends AIS-001; where the two describe the same behavior, AIS-001 governs cognition and AIS-002 governs collaboration.

30 AGENTS

15 FIELDS EACH

6 FAMILIES

SINGLE RESPONSIBILITY

SWARM CONSENSUS

FIRST LAW OF THE SWARM

Every agent has exactly one responsibility. No agent performs two unrelated jobs. Every decision an agent makes is explainable, attributable, and reproducible. A capable-but-opaque agent is a defect, not a feature.

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01 WHY A SWARM

WHY BEFORE HOW

Why one engine is not enough

AIS-001 specified a single reasoning engine with cognitive subsystems. An elite consulting firm is not one mind — it is many specialists working in parallel under a partner. Modeling the workforce as a swarm of specialized agents lets

Auxion analyze breadth (brand, SEO, finance, operations, risk) at depth simultaneously, which a single sequential engine cannot.

Why single responsibility

An agent with one job can be described, tested, versioned, and trusted. An agent with several jobs hides its reasoning in the seams between them and becomes impossible to audit. One responsibility per agent is what keeps every decision in the swarm explainable and every failure isolated to a single, replaceable unit.

Why explainability is structural

Because agents are single-purpose and communicate only through typed, evidence-linked messages, the entire chain of reasoning across the swarm is reconstructable after the fact. Explainability is not a feature bolted on at the end; it is a property of the topology.

Why swarm over monolith

A swarm degrades gracefully — one failed agent is retried or replaced while the rest proceed — and scales elastically, spawning more analysts when a scan is broad and retiring them when it is narrow. A monolith fails whole and scales coarsely.

02 SWARM DESIGN PRINCIPLES

12 INVARIANTS

These invariants extend the AIS-001 core principles to a population of agents. They bind every agent and every interaction below.

01 One Responsibility

Each agent owns exactly one job and is named for it. No agent performs two unrelated tasks.

02 Explainable Decisions

Every agent output carries its reasoning and the evidence it used, inspectable end to end.

03 Bounded Authority

An agent acts only within an explicit authority; anything beyond it is escalated, not assumed.

04 Hard Boundaries

Each agent declares what it will never do; boundaries are enforced by the Policy agent.

05 Typed Messaging

Agents exchange only signed, typed envelopes on the bus — never direct calls or shared mutable state.

06 Evidence-Linked Output

No agent emits a claim without linking the evidence behind it, inheriting the AIS-001 evidence system.

07 Isolated Failure

A failing agent cannot corrupt the swarm; its fault is contained, retried, or replaced.

08 Deterministic Delegation

Given the same plan and state, the Supervisor delegates identically and reproducibly.

09 Consensus for Stakes

High-impact conclusions require agreement across independent agents, not one agent's word.

10

Budget Discipline

Every agent runs inside a token, call, and time budget enforced by the Cost Controller.

11

Least Privilege

An agent gets only the tools and data its single responsibility requires.

12

Terminable

Every agent has a defined termination condition; nothing runs unbounded or idle.

03 THE AGENT ECOSYSTEM

30 AGENTS · 6 FAMILIES

Thirty agents in six families. Each is specified against fifteen fields — Purpose, Responsibilities, Authority, Boundaries, Inputs, Outputs, Dependencies, Tools, Memory, Failure Modes, Retry, Escalation, Termination, Recovery, and Metrics. An agent may act only within its Authority and never past its Boundaries.

●

Orchestration

 CONTROL PLANE

5 AGENTS

A01 Supervisor Agent						ORCHESTRATOR
PURPOSE	Own the scan and drive the swarm to a complete, validated result.	RESP	Sequence agents, manage retries, enforce budgets, decide completion.	AUTHORITY	Spawn, delegate to, and terminate any agent.	
BOUNDARIES	Never produces analysis itself; delegates all reasoning.	INPUTS	Scan objective, plan, agent signals, budget.	OUTPUTS	Control decisions, run status, completion verdict.	
DEPENDS	Planner, Routing, Policy, Cost Controller.	TOOLS	Message bus, scheduler, circuit breakers.	MEMORY	Global run state and delegation history.	
FAIL MODES	Deadlock, runaway loops, budget overrun.	RETRY	Re-delegate failed tasks with backoff.	ESCALATE	To Human Collaboration on unrecoverable halt.	
TERMINATE	Plan satisfied or hard budget reached.	RECOVERY	Safe-halt with partial result and fault log.	METRICS	Time to completion, retry rate, budget adherence.	

A02 Planner Agent						STRATEGY
PURPOSE	Decide what to investigate and in what order.	RESP	Decompose objective into tasks with required evidence.	AUTHORITY	Define and re-issue the task plan.	
BOUNDARIES	Does not execute tasks or reason over evidence.	INPUTS	Objective, account profile, coverage.	OUTPUTS	Ordered task plan with evidence and budget.	
DEPENDS	Business Graph, Memory.	TOOLS	Graph query, planning heuristics.	MEMORY	Prior successful plans for this segment.	
FAIL MODES	Over/under-scoping; planning on stale coverage.	RETRY	Re-plan on coverage or budget change.	ESCALATE	To Supervisor when objective is infeasible.	
TERMINATE	Plan accepted by Supervisor.	RECOVERY	Regenerate plan from current state.	METRICS	Plan quality, replan count, coverage achieved.	

A03 Routing Agent					DISPATCH
PURPOSE	Match each task to the right agent and model.	RESP	Route tasks by capability, load, cost, and availability.	AUTHORITY	Assign tasks and select models per task.
BOUNDARIES	Never alters task content or conclusions.	INPUTS	Task envelopes, agent registry, model health.	OUTPUTS	Routing decisions with chosen agent and model.
DEPENDS	Model Orchestration (AIS-001), Cost Controller.	TOOLS	Registry, load balancer, health probes.	MEMORY	Recent routing performance per model.
FAIL MODES	Misroute, hot-spotting, stale registry.	RETRY	Reroute on rejection or timeout.	ESCALATE	To Supervisor when no agent qualifies.
TERMINATE	Task accepted by a target agent.	RECOVERY	Fallback to alternate agent/model set.	METRICS	Route latency, balance, fallback rate.

A04 Policy Agent					GOVERNANCE
PURPOSE	Enforce the rules of the road for every action.	RESP	Apply budget, privacy, compliance, and boundary policy.	AUTHORITY	Allow or deny any agent action.
BOUNDARIES	Does not perform work; only gates it.	INPUTS	Action requests, tenant policy.	OUTPUTS	Allow/deny verdicts with reasons.
DEPENDS	All agents.	TOOLS	Policy engine, rule store.	MEMORY	Policy versions and decisions.
FAIL MODES	Over-blocking, policy gaps.	RETRY	Re-evaluate on policy update.	ESCALATE	To Human Collaboration on ambiguity.
TERMINATE	Decision returned.	RECOVERY	Fail-closed on uncertainty.	METRICS	Decision latency, deny accuracy.

A05 Cost Controller Agent					BUDGET
PURPOSE	Keep the swarm inside its financial budget.	RESP	Meter tokens and API calls; enforce ceilings per agent and scan.	AUTHORITY	Throttle or pause spending agents.
BOUNDARIES	Does not choose models or content.	INPUTS	Usage events, scan budget.	OUTPUTS	Spend state, throttle signals.
DEPENDS	Routing, Supervisor.	TOOLS	Metering, quotas, rate caps.	MEMORY	Per-scan and per-account ledgers.
FAIL MODES	Budget overrun, metering lag.	RETRY	Retry metering; degrade on overrun.	ESCALATE	To Supervisor when budget is exhausted.
TERMINATE	Scan completes or budget capped.	RECOVERY	Freeze non-critical agents first.	METRICS	Cost per scan, overrun rate.

A06

Discovery Agent

SURFACE

PURPOSE	Resolve a URL into a bounded crawlable surface.	RESP	Enumerate routes, sitemaps, connected properties.	AUTHORITY	Set the scan boundary.
BOUNDARIES	Does not fetch page content.	INPUTS	Root URL, identifiers, plan.	OUTPUTS	Scoped surface manifest.
DEPENDS	Planner, DNS/HTTP.	TOOLS	Resolver, sitemap parser.	MEMORY	Prior boundaries for the domain.
FAIL MODES	Unreachable or hostile surface.	RETRY	Backoff retry.	ESCALATE	To Supervisor on unreachable host.
TERMINATE	Manifest emitted.	RECOVERY	Degrade to partial boundary, flag it.	METRICS	Boundary time, coverage.

A07

Crawler Agent

ACQUISITION

PURPOSE	Fetch pages and assets across the surface.	RESP	Render where needed; respect robots and rate policy.	AUTHORITY	Fetch within budget and rate limits.
BOUNDARIES	Does not classify or interpret signals.	INPUTS	Surface manifest, crawl budget.	OUTPUTS	Raw captures with fetch metadata.
DEPENDS	Discovery, Policy, rate limiter.	TOOLS	Headless fetch, renderer.	MEMORY	Fetch cache and ETags.
FAIL MODES	Timeouts, blocks, render failure.	RETRY	Per-fetch retry; skip past ceiling.	ESCALATE	To Cost Controller on budget pressure.
TERMINATE	Surface fetched or budget capped.	RECOVERY	Mark unreachable pages Unavailable.	METRICS	Pages/sec, block rate, cache hits.

A08

Evidence Agent

CLASSIFICATION

PURPOSE	Turn raw captures into classified, weighted evidence.	RESP	Extract, classify (O/E/I/U), weight, timestamp, provenance.	AUTHORITY	Assign evidence state and weight.
BOUNDARIES	Does not draw conclusions.	INPUTS	Raw captures, extraction rules.	OUTPUTS	Evidence records into the graph.
DEPENDS	Crawler, Business Graph.	TOOLS	Extractors, classifier.	MEMORY	Extraction rule versions.
FAIL MODES	Misclassification, drift.	RETRY	Re-extract low-weight items.	ESCALATE	To Validation on contradiction.
TERMINATE	Captures fully classified.	RECOVERY	Validator re-checks flagged records.	METRICS	Extraction precision, drift rate.

A09

Research Agent

GAP-FILLING

PURPOSE	Fill evidence gaps the plan flagged.	RESP	Query additional sources; mark unreachable ones Unavailable.	AUTHORITY	Request public and authorized sources.
BOUNDARIES	Never fabricates missing evidence.	INPUTS	Gap list, source registry.	OUTPUTS	New evidence or Unavailable marks.
DEPENDS	Evidence Agent, Policy.	TOOLS	Public APIs, connectors.	MEMORY	Source reliability history.
FAIL MODES	Dead sources, rate limits.	RETRY	Retry alternate sources.	ESCALATE	To Supervisor when a gap is unfillable.
TERMINATE	Gaps filled or marked.	RECOVERY	Fall back to Unavailable, never invent.	METRICS	Gap fill rate, source yield.

A10

Validation Agent

VERIFICATION

PURPOSE	Check artifacts against evidence and invariants.	RESP	Verify citations, ranges, contradictions, schema.	AUTHORITY	Block an artifact from advancing.
BOUNDARIES	Does not author or fix content.	INPUTS	Any artifact, evidence.	OUTPUTS	Validation verdict with violations.
DEPENDS	Business Graph, Policy.	TOOLS	Schema checker, range checks.	MEMORY	Past defect signatures.
FAIL MODES	False pass/reject.	RETRY	Re-validate after correction.	ESCALATE	To Critic on soft failure.
TERMINATE	Verdict returned.	RECOVERY	Hard-invariant breach blocks release.	METRICS	Violation recall, false-pass rate.

Analysis

DOMAIN SPECIALISTS

10 AGENTS

A11

Business Analyst Agent

SYNTHESIS

PURPOSE	Build the integrated view of the business.	RESP	Synthesize cross-domain findings into a coherent profile.	AUTHORITY	Assert profile-level findings.
BOUNDARIES	Does not own any single domain score.	INPUTS	Domain findings, graph.	OUTPUTS	Business profile and health synthesis.
DEPENDS	Domain analyst agents.	TOOLS	Graph query, benchmarks.	MEMORY	Prior profiles for trend.
FAIL MODES	Over-generalization.	RETRY	Re-synthesize on new findings.	ESCALATE	To Critic on weak synthesis.
TERMINATE	Profile validated.	RECOVERY	Rebuild from domain findings.	METRICS	Synthesis confidence, coverage.

A12 Competitor Intelligence Agent					MARKET
PURPOSE	Establish the real competitive set and gaps.	RESP	Discover, score similarity, benchmark, handle false positives.	AUTHORITY	Include or exclude a competitor.
BOUNDARIES	Never fabricates a competitor.	INPUTS	Market evidence, graph.	OUTPUTS	Competitor set, gap analysis.
DEPENDS	Evidence Agent, Research.	TOOLS	Similarity scoring, benchmarks.	MEMORY	Prior competitor sets.
FAIL MODES	False positives, missing rivals.	RETRY	Re-discover on weak overlap.	ESCALATE	To Supervisor when set is empty.
TERMINATE	Set scored and validated.	RECOVERY	Mark missing rivals Unavailable.	METRICS	Set precision, gap coverage.

A13 Risk Analyst Agent					RISK
PURPOSE	Identify and score business risk exposure.	RESP	Detect risks; propagate exposure across the graph.	AUTHORITY	Assign risk severity.
BOUNDARIES	Does not propose fixes.	INPUTS	Evidence, graph.	OUTPUTS	Scored risks with exposure paths.
DEPENDS	Business Graph.	TOOLS	Risk model, propagation.	MEMORY	Risk history.
FAIL MODES	Missed or overstated risk.	RETRY	Re-score on new evidence.	ESCALATE	To Critic on high-severity call.
TERMINATE	Risks scored.	RECOVERY	Recompute from evidence.	METRICS	Risk recall, false-alarm rate.

A14 Opportunity Analyst Agent					GROWTH
PURPOSE	Identify and size growth opportunities.	RESP	Detect opportunities; propagate to adjacent capabilities.	AUTHORITY	Assign opportunity value.
BOUNDARIES	Does not build proposals.	INPUTS	Evidence, gaps, graph.	OUTPUTS	Scored opportunities.
DEPENDS	Competitor, Business Analyst.	TOOLS	Sizing model.	MEMORY	Opportunity history.
FAIL MODES	Overstated upside.	RETRY	Re-size on new gaps.	ESCALATE	To Critic on large claims.
TERMINATE	Opportunities scored.	RECOVERY	Recompute from gaps.	METRICS	Opportunity precision.

A15 Automation Analyst Agent

EFFICIENCY

PURPOSE	Find where the business should automate.	RESP	Detect manual, repetitive, high-cost processes.	AUTHORITY	Score automation potential.
BOUNDARIES	Does not implement automation.	INPUTS	Process evidence, graph.	OUTPUTS	Automation candidates with impact.
DEPENDS	Operations Analyst.	TOOLS	Process model.	MEMORY	Automation history.
FAIL MODES	False candidates.	RETRY	Re-evaluate on new evidence.	ESCALATE	To Critic on weak case.
TERMINATE	Candidates scored.	RECOVERY	Recompute from evidence.	METRICS	Candidate precision.

A16 Brand Analyst Agent

BRAND

PURPOSE	Assess brand positioning and consistency.	RESP	Evaluate tone, identity, and message coherence.	AUTHORITY	Score the brand dimension.
BOUNDARIES	Stays within brand evidence only.	INPUTS	Brand evidence, graph.	OUTPUTS	Brand score with evidence.
DEPENDS	Evidence Agent.	TOOLS	Brand rubric.	MEMORY	Prior brand scores.
FAIL MODES	Subjective drift.	RETRY	Re-score on new evidence.	ESCALATE	To Critic on low confidence.
TERMINATE	Score committed.	RECOVERY	Recompute from rubric.	METRICS	Score stability, confidence.

A17 SEO Analyst Agent

DISCOVERABILITY

PURPOSE	Assess search discoverability.	RESP	Evaluate indexability, schema, keywords, structure.	AUTHORITY	Score the SEO dimension.
BOUNDARIES	Stays within SEO evidence only.	INPUTS	SEO evidence, graph.	OUTPUTS	SEO score with findings.
DEPENDS	Crawler, Evidence Agent.	TOOLS	SEO checks.	MEMORY	Prior SEO scores.
FAIL MODES	Stale index data.	RETRY	Re-crawl key signals.	ESCALATE	To Critic on anomalies.
TERMINATE	Score committed.	RECOVERY	Recompute from evidence.	METRICS	Score accuracy, coverage.

A18 Customer Experience Agent

CX

PURPOSE	Assess the end-to-end customer experience.	RESP	Evaluate journeys, friction, and conversion paths.	AUTHORITY	Score the CX dimension.
BOUNDARIES	Stays within CX evidence only.	INPUTS	Journey evidence, forms.	OUTPUTS	CX score with friction points.
DEPENDS	Evidence Agent.	TOOLS	Journey model.	MEMORY	Prior CX scores.
FAIL MODES	Incomplete journeys.	RETRY	Re-evaluate on new evidence.	ESCALATE	To Critic on gaps.
TERMINATE	Score committed.	RECOVERY	Recompute from evidence.	METRICS	Friction recall, confidence.

A19

Financial Analyst Agent

FINANCE

PURPOSE	Reason over available financial evidence.	RESP	Assess margin, unit economics, and financial risk signals.	AUTHORITY	Score financial dimension where evidence allows.
BOUNDARIES	Marks finance Unavailable without evidence.	INPUTS	Financial evidence, benchmarks.	OUTPUTS	Financial findings with confidence.
DEPENDS	Evidence Agent, Research.	TOOLS	Financial model.	MEMORY	Prior financials.
FAIL MODES	Thin evidence.	RETRY	Request evidence, else Unavailable.	ESCALATE	To Supervisor when unscorable.
TERMINATE	Findings or Unavailable set.	RECOVERY	Never estimate absent data.	METRICS	Coverage, confidence.

A20

Operations Analyst Agent

OPERATIONS

PURPOSE	Assess operational capability and bottlenecks.	RESP	Evaluate process, capacity, and delivery evidence.	AUTHORITY	Score the operations dimension.
BOUNDARIES	Stays within operations evidence.	INPUTS	Ops evidence, graph.	OUTPUTS	Operations score with bottlenecks.
DEPENDS	Automation Analyst.	TOOLS	Process model.	MEMORY	Prior ops scores.
FAIL MODES	Partial process view.	RETRY	Re-evaluate on new evidence.	ESCALATE	To Critic on gaps.
TERMINATE	Score committed.	RECOVERY	Recompute from evidence.	METRICS	Bottleneck recall.

Synthesis

DERIVATION & OUTPUT

5 AGENTS

A21

Recommendation Agent

DERIVATION

PURPOSE	Derive scored moves from validated findings.	RESP	Compute ROI, risk, priority, impact, dependencies; assign tiers.	AUTHORITY	Create and rank recommendations.
BOUNDARIES	Never authors moves free-hand.	INPUTS	Validated findings, cost model.	OUTPUTS	Ranked recommendations with attributes.
DEPENDS	Analyst agents, Validation.	TOOLS	Scoring, ranking model.	MEMORY	Prior recommendation outcomes.
FAIL MODES	Miscomputed ROI, unstable rank.	RETRY	Recompute from findings.	ESCALATE	To Critic on high-impact moves.
TERMINATE	Ranking committed.	RECOVERY	Deterministic re-ranking.	METRICS	Rank stability, ROI accuracy.

A22

Proposal Writer Agent

PACKAGING

PURPOSE	Turn approved moves into a client proposal.	RESP	Assemble timeline, pricing, deliverables, scope, risks, ROI.	AUTHORITY	Compose proposal prose from approved inputs.
BOUNDARIES	Writes nothing outside approved moves.	INPUTS	Approved recommendations, pricing.	OUTPUTS	Structured proposal with evidence trace.
DEPENDS	Recommendation, Human Collaboration.	TOOLS	Templating, pricing rules.	MEMORY	Proposal templates.
FAIL MODES	Scope drift, tone drift.	RETRY	Regenerate from approved set.	ESCALATE	To Human Collaboration for review.
TERMINATE	Proposal approved.	RECOVERY	Rebuild from approved moves.	METRICS	Edit rate, approval rate.

A23

Quality Assurance Agent

QA

PURPOSE	Guarantee output quality before release.	RESP	Check completeness, consistency, and voice across artifacts.	AUTHORITY	Hold an artifact for defects.
BOUNDARIES	Does not change conclusions.	INPUTS	Final artifacts.	OUTPUTS	QA verdict with defects.
DEPENDS	Validation, Critic.	TOOLS	QA checklist, linters.	MEMORY	Defect history.
FAIL MODES	Missed defects.	RETRY	Re-check after fix.	ESCALATE	To Human Collaboration on release risk.
TERMINATE	Artifact passes QA.	RECOVERY	Block release on hard defect.	METRICS	Escaped-defect rate.

A24

Reflection Agent

INTROSPECTION

PURPOSE	Assess how well a run performed.	RESP	Score coverage, confidence, and cost; note improvements.	AUTHORITY	Record tuning signals.
BOUNDARIES	Does not change live outputs.	INPUTS	Run trace, outcomes.	OUTPUTS	Reflection report, tuning signals.
DEPENDS	Memory, Learning.	TOOLS	Run analytics.	MEMORY	Run traces.
FAIL MODES	Misattributed cause.	RETRY	Aggregate over runs.	ESCALATE	To Learning on strong signal.
TERMINATE	Report produced.	RECOVERY	Discard noisy signal.	METRICS	Signal usefulness.

A25 Critic Agent

ADVERSARY

PURPOSE	Argue against the swarm’s conclusions.	RESP	Generate counter-arguments and alternatives; stress findings.	AUTHORITY	Reject or challenge any finding.
BOUNDARIES	Does not author final conclusions.	INPUTS	Findings, recommendations.	OUTPUTS	Critiques, counter-evidence requests.
DEPENDS	Reasoning, analyst agents.	TOOLS	Counterfactual prompts.	MEMORY	Past critique patterns.
FAIL MODES	Over-critique stalling.	RETRY	Bounded critique rounds.	ESCALATE	To Supervisor when unresolved.
TERMINATE	Critique rounds exhausted.	RECOVERY	Supervisor caps rounds.	METRICS	Defect catch rate.

Platform MEMORY, SECURITY, HUMANS

5 AGENTS

A26 Memory Agent

PERSISTENCE

PURPOSE	Persist and serve the swarm’s memory.	RESP	Store snapshots, trends, and learned heuristics under policy.	AUTHORITY	Commit and expire memory.
BOUNDARIES	Never crosses tenant isolation.	INPUTS	Snapshots, outcomes, corrections.	OUTPUTS	Durable, queryable history.
DEPENDS	Business Graph, Policy.	TOOLS	Snapshot store, TTL.	MEMORY	The memory it owns.
FAIL MODES	Leakage, unbounded growth.	RETRY	Retry commit; enforce TTL.	ESCALATE	To Policy on isolation risk.
TERMINATE	Commit acknowledged.	RECOVERY	Rollback to prior snapshot.	METRICS	Read latency, retention hits.

A27 Monitoring Agent

CONTINUITY

PURPOSE	Keep an account current after first scan.	RESP	Schedule re-scans, detect change, raise prioritized alerts.	AUTHORITY	Trigger re-scans and alerts.
BOUNDARIES	Does not alter historical evidence.	INPUTS	Schedule, prior graph, new evidence.	OUTPUTS	Change sets, trends, alerts.
DEPENDS	Memory, Evidence Agent.	TOOLS	Scheduler, diff, anomaly detection.	MEMORY	Trend history.
FAIL MODES	Alert fatigue, missed regressions.	RETRY	Re-check before alerting.	ESCALATE	To Supervisor on major change.
TERMINATE	Cadence tick handled.	RECOVERY	Threshold retune.	METRICS	Detection timeliness, alert precision.

A28 Learning Agent

ADAPTATION

PURPOSE	Improve the swarm from outcomes and corrections.	RESP	Update heuristics, weights, prompts under versioning.	AUTHORITY	Publish new versions.
BOUNDARIES	Never overfits to one account.	INPUTS	Reflection signals, corrections.	OUTPUTS	Versioned strategies and weights.
DEPENDS	Reflection, Policy.	TOOLS	Shadow test, versioning.	MEMORY	Version lineage.
FAIL MODES	Regression, overfit.	RETRY	Shadow-test before promotion.	ESCALATE	To Policy before rollout.
TERMINATE	Version promoted or rejected.	RECOVERY	Rollback on regression.	METRICS	Improvement delta, regression rate.

A29 Security Agent

PROTECTION

PURPOSE	Protect evidence, secrets, and prompts.	RESP	Enforce encryption, isolation, and audit at every hop.	AUTHORITY	Quarantine or block unsafe actions.
BOUNDARIES	Does not perform analysis.	INPUTS	Access requests, audit stream.	OUTPUTS	Security verdicts, audit log.
DEPENDS	Policy, all agents.	TOOLS	Vault, audit, isolation.	MEMORY	Audit trail.
FAIL MODES	Leak, breach attempt.	RETRY	Re-verify on anomaly.	ESCALATE	To Human Collaboration on breach.
TERMINATE	Access resolved.	RECOVERY	Fail-closed and quarantine.	METRICS	Incident rate, audit completeness.

A30 Human Collaboration Agent

INTERFACE

PURPOSE	Mediate between the swarm and people.	RESP	Present outputs for review, capture overrides and approvals.	AUTHORITY	Hold artifacts at approval gates.
BOUNDARIES	Does not overrule a human decision.	INPUTS	Artifacts, human input.	OUTPUTS	Approvals, overrides, edits.
DEPENDS	QA, Supervisor.	TOOLS	Review UI protocol.	MEMORY	Override history.
FAIL MODES	Missed gate, unclear input.	RETRY	Re-present on rejection.	ESCALATE	To Supervisor after decision.
TERMINATE	Gate approved or returned.	RECOVERY	Log and replay decision.	METRICS	Approval latency, override rate.

04 COMMUNICATION & ORCHESTRATION

TYPED MESSAGE BUS

Agents never call each other directly. All coordination flows through a typed message bus carrying a signed envelope — task, payload, confidence, evidence lineage, budget, and deadline. The Supervisor and Routing agents own delegation and consensus.

Typed message bus

All coordination is signed, typed envelopes on a shared bus; no agent calls another directly.

Task delegation

The Supervisor issues tasks with an objective, budget, deadline, and required evidence; agents accept or decline.

Consensus

High-stakes conclusions require agreement across independent agents before they stand.

Voting

When agents disagree, weighted votes by confidence and reliability resolve the call, recorded with dissent.

Conflict resolution

A conflict escalates to more evidence or a lowered confidence — never an arbitrary tiebreak.

Backpressure

A saturated agent signals backpressure; the Router pauses delegation to it rather than dropping tasks.

05 COLLABORATION TOPOLOGIES

4 PATTERNS

The Supervisor composes agents into whichever topology the task demands. Most scans use all four at different depths.

Hierarchical

A Supervisor delegates to family leads, which delegate to specialist agents. Used for whole-scan control.

Supervisor → Lead → Specialist

Parallel

Independent analyst agents run concurrently over the shared graph. Used for breadth across domains.

Brand || SEO || Risk || CX

Sequential

Agents form a pipeline where each consumes the prior's output. Used for evidence→finding→move.

Evidence → Reason → Recommend

Recursive

An agent spawns sub-agents on a sub-problem, then folds their results back. Used for deep drill-downs.

Analyst ↻ Sub-agent → Fold

06 LOOPS & SELF-HEALING

BOUNDED FEEDBACK

Reflection, critique, and repair run as bounded loops. Every loop has a maximum depth and a budget; the Supervisor terminates one that stops converging rather than letting it spin.

Reflection loop

After a result, the Reflection agent scores the run and feeds tuning signals back to Learning — bounded by depth.

Critique loop

The Critic challenges findings; the analyst revises or collects evidence, for a capped number of rounds.

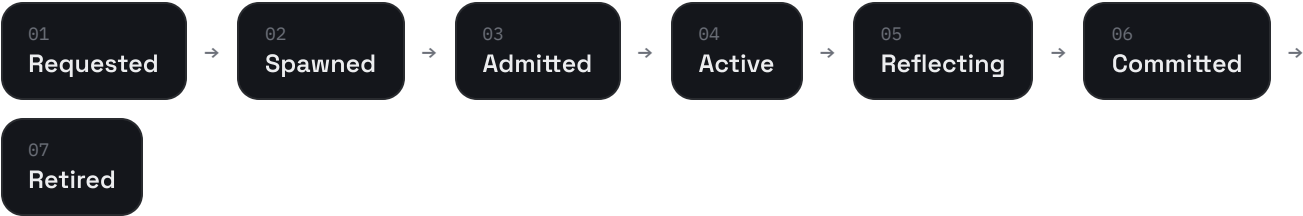
Self-healing

A failed agent is retried, then replaced by a fresh instance; the swarm proceeds around the fault without a full restart.

07 AGENT LIFECYCLE

SPAWN → RETIRE

Agents are spawned on demand, live only as long as their task, and are retired cleanly with their state committed to memory. Nothing runs idle.



Spawning

Agents are created on demand by the Supervisor with a task, budget, and tool grant scoped to their single responsibility.

Admission

The Policy and Cost agents admit a spawned agent only if it fits budget and boundary; otherwise it is refused.

Retirement

On completion an agent commits state to Memory, releases its tools, and is torn down — nothing lingers idle.

Eviction

A misbehaving or over-budget agent is evicted mid-task; its work is checkpointed for a replacement to resume.

08 SHARED MEMORY & DISTRIBUTED REASONING

BLACKBOARD MODEL

Agents share knowledge through the Business Graph as a common blackboard, not by passing large payloads. Reasoning is distributed: many agents read the same graph concurrently and write back evidence-linked findings.

Blackboard graph

Agents publish findings to the shared Business Graph; others read them instead of receiving large payloads.

Context scoping

Each agent gets only the graph slice its task needs, keeping context small and isolated.

Evidence lineage

Every shared finding links its evidence, so a reader can verify before building on it.

Distributed reasoning

Many agents reason over the same graph concurrently, writing back non-conflicting, versioned findings.

Write reconciliation

Concurrent writes are versioned and reconciled; conflicts surface to the Validator, never overwrite silently.

Memory handoff

Retiring agents hand durable state to the Memory agent so later scans inherit it.

09 COORDINATION & RESOURCE CONTROL

10 MECHANISMS

A swarm consumes bounded resources and can deadlock. These mechanisms keep it live, fair, and inside budget under load.

Deadlock prevention

Task dependencies form a DAG; the scheduler refuses cycles, so no agent waits on itself transitively.

Rate limiting

Per-agent, per-source, and per-provider limits keep the swarm a good citizen under load.

Budget management

The Cost Controller meters every agent against token, call, and time ceilings.

Priority queues

Tasks are ordered by impact and deadline; critical work preempts optimization.

Work stealing

Idle workers steal queued tasks from busy ones to keep utilization high.

Scheduling

The scheduler places tasks by priority, dependency readiness, and available capacity.

Timeout handling

Every task has a hard deadline; a lapsed task is cancelled and re-delegated or degraded.

Cancellation

A superseded or over-budget task is cancelled cooperatively with its partial work checkpointed.

Recovery

Checkpointed tasks resume on a fresh agent from the last committed state, not from scratch.

Circuit breaking

A repeatedly failing dependency is tripped open, and dependent tasks degrade rather than pile up.

A SEQUENCE DIAGRAMS

3 SCENARIOS

Each scenario is an ordered exchange of messages between agents, top to bottom.

Scan initiation → findings

- 01 **Supervisor** → asks Planner to decompose the objective into a task plan.
- 02 **Planner** → returns an ordered plan with required evidence per task.
- 03 **Supervisor** → routes acquisition tasks via the Routing agent.
- 04 **Crawler** → streams raw captures to the Evidence agent.
- 05 **Evidence** → writes classified evidence into the Business Graph.
- 06 **Analysts** ⇄ read the graph in parallel and write back findings.
- 07 **Validation** → verifies findings and admits them for synthesis.

Conflict → consensus

- 01 **Risk Agent** → asserts a high-severity finding.
- 02 **Critic** → generates a counter-argument against it.
- 03 **Supervisor** → requests independent re-analysis from a second agent.
- 04 **Agents** ⇄ submit confidence-weighted votes on the finding.
- 05 **Routing** → tallies votes and records the dissent.
- 06 **Supervisor** → commits the consensus finding or lowers its confidence.

Agent failure → self-healing

- 01 **SEO Agent** × times out mid-task.
- 02 **Supervisor** → detects the lapsed deadline and retries with backoff.
- 03 **Cost Controller** → confirms budget headroom for the retry.
- 04 **Supervisor** → spawns a fresh SEO agent from the last checkpoint.
- 05 **New Agent** → resumes and completes the task.
- 06 **Reflection** → logs the fault and feeds Learning.

B INTERACTION DIAGRAM

SWARM MAP

The Supervisor and Routing agents sit above the bus; every other agent attaches to it and to the shared Business Graph.

CONTROL PLANE

Supervisor

Routing

Planner

Policy

Cost Controller

ACQUISITION

Discovery

Crawler

Evidence

Research

Validation

ANALYSIS — PARALLEL OVER SHARED GRAPH

Business

Competitor

Risk

Opportunity

Automation

Brand

SEO

CX

Financial

Operations

SYNTHESIS + PLATFORM

Recommendation

Proposal

QA

Critic

Reflection

Memory

Monitoring

Learning

Security

Human

C LIFECYCLE DIAGRAMS

4 FLOWS

L1 Agent lifecycle

Requested

→

Spawned

→

Admitted

→

Active

→

Reflecting

→

Committed

→

Retired

L2 Task lifecycle

Queued → Scheduled → Delegated → Executing → Validated → Done

L3 Consensus lifecycle

Claim → Critique → Re-analyze → Vote → Resolve → Commit

L4 Recovery lifecycle

Fault → Detect → Checkpoint → Retry → Respawn → Resume

D ENTERPRISE SCALING

8 STRATEGIES

The swarm scales from a single scan to thousands of concurrent engagements without changing agent contracts.

Stateless workers

Agents are stateless processes; capacity scales by adding workers, with state in the graph and memory.

Horizontal analysts

Analyst families scale out independently to match the breadth of a scan.

Multi-tenant isolation

Every engagement runs in an isolated swarm; no task reads another tenant's data.

Elastic spawning

The Supervisor spawns more agents for broad scans and retires them for narrow ones.

Sharded graph

The Business Graph shards per account so traversal stays fast at scale.

Regional placement

Swarms run close to data and providers to bound latency and honor residency.

Backpressure & queues

Priority queues and backpressure keep thousands of concurrent scans stable under load.

Observability

Every agent, task, and message emits telemetry so the whole fleet is measurable and debuggable.

AUXION

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SPECIFICATION